

Joren Gijsbrechts

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SUMMARY

As an Assistant Professor of Operations Management at Esade Business School, I create algorithms that translate complex datasets into actionable decisions. I also deliver custom talks on AI, analytics, and Supply Chain 4.0.

EDUCATION

University of Antwerp

B.Sc. & M.Sc. Applied Economics

Antwerp, Belgium

Sep. 2009 – June 2012

KU Leuven

Ph.D. Operations Management

Leuven, Belgium

Sep. 2016 – August 2020

ACADEMIC EXPERIENCE

Assistant Professor

Esade Business School, Ramon Llull University

Sep. 2023 – Present

Barcelona, Spain

- Teaching core Operations and Business Analytics courses.

Assistant Professor

Catolica Lisbon School of Business and Economics

Sep. 2020 – Aug. 2023

Lisbon, Portugal

- Taught core Operations and Business Analytics courses.

Visiting Scholar

MIT Sloan School of Management

Jan. 2023 – March. 2023

Cambridge, Massachusetts

- Joint Research with professor Vivek Farias.

Visiting PhD student

Kellogg School of Management

Sep. 2019 – Dec. 2019

Evanston, Illinois

- Joint Research with professor Jan A. Van Mieghem.

INDUSTRY EXPERIENCE

Advisor, Algorithms

CurveCatch

March 2020 – present

Antwerp, Belgium

- CurveCatch is a start-up disrupting the lingerie industry (www.curvecatch.com).
- I collaborate with the founding team to design and implement algorithms.

Supply Chain Analyst

Procter and Gamble

June. 2014 – Aug. 2016

Stockholm, Sweden

- Prior to starting my Ph.D., I worked two years in industry.
- Analyses as part of monthly Sales & Operations planning cycle in Nordic market.

PUBLICATIONS

Journal Articles

- **Gijsbrechts J, Van Staden HE.** Where Did the Demand Go? Teaching Demand Censoring with the Newsvendor Challenge. *INFORMS Transactions on Education* (2024). DOI:10.1287/ited.2024.0085
- **Gijsbrechts J, Imdahl C, Boute RN, Van Mieghem J.** Optimal Robust Inventory Management with Volume Flexibility: Matching Capacity and Demand with the Lookahead Peak-Shaving Policy. *Production and Operations Management* 32(11):3357–3373 (2023). DOI:10.1111/poms.14069
- **Gijsbrechts J, Boute RN, Disney SM, Van Mieghem JA.** Volume Flexibility at Responsive Suppliers in Reshoring Decisions: Analysis of a Dual Sourcing Inventory Model. *Production and Operations Management* (2022). DOI:10.1111/poms.13719

- **Gijsbrechts J, Boute RN, Van Mieghem JA, Zhang DJ.** Can Deep Reinforcement Learning Improve Inventory Management? Performance on Dual Sourcing, Lost Sales and Multi-Echelon Problems. *Manufacturing & Services Operations Management* 24(3):1261–1885 (2022). DOI:10.1287/msom.2021.1064
- **Boute RN, Disney SM, Gijsbrechts J, Van Mieghem JA.** Dual Sourcing and Smoothing under Non-Stationary Demand Time Series: Re-shoring with SpeedFactories. *Management Science* 68(2):1039–1057 (2021). DOI:10.1287/mnsc.2020.3951
- **De Moor B, Boute RN, Gijsbrechts J.** A reward shaping approach to improve the applicability of deep reinforcement learning to inventory management. *European Journal of Operational Research* 301(2):535–545 (2021). DOI:10.1016/j.ejor.2021.10.045
- **Boute RN, Gijsbrechts J, van Jaarsveld W, Vanvuchelen N.** Deep reinforcement learning for inventory control: A roadmap. *European Journal of Operational Research* 298(2):401–412. DOI:10.1016/j.ejor.2021.07.016
- **Yee H, Boute RN, Gijsbrechts J.** Synchromodal Planning under Stochastic Travel Times. *Computers in Industry* 125:103367 (2021). DOI:10.1016/j.compind.2020.103367
- **Vanvuchelen N, Gijsbrechts J, Boute RN.** Use of Proximal Policy Optimization for the Joint Replenishment Problem. *Computers in Industry* 119:103239 (2020). DOI:10.1016/j.compind.2020.103239

Working Papers

- **Farias V, Gijsbrechts J, Khojandi A, Peng T, Zheng A.** Speeding up Policy Simulation in Supply Chain RL. Working paper at MIT. arXiv:2406.01939
- **Bertani N, Gijsbrechts J, Godinho de Matos M, Wagner L.** Reminders in a Resource Scarce Setting: The Case of Mobile Data Consumption. Working paper at Católica Lisbon School of Business & Economics. <https://ssrn.com/abstract=4911622>
- **De Moor B, Boute RN, Gijsbrechts J.** When do Shippers Match? Managing Inventory with Capacitated Discount Opportunities. Paper available upon request.
- **Gijsbrechts J, Olivares Nadal A, Wagner L.** Replenishing a Try-Before-You-Buy Channel in Online Retail. Working paper at Esade, Ramon Llull University.

Book Chapters

- **Boute RN, Gijsbrechts J, Van Mieghem JA.** Digital Lean Operations: Smart Automation and Artificial Intelligence in Financial Services. In: Babich V, Birge J, Hilary G (eds) *Innovative Technology at the interface of Finance and Operations*. Springer Series in Supply Chain Management, Springer Nature (2021).
- **Gijsbrechts J, Boute RN, Van Mieghem JA, Zhang D.** AI in Inventory Management: The Disruptive Era of DRL and Beyond. Forthcoming chapter in “AI in Supply Chains: Perspectives from Global Thought Leaders,” edited by Maxime C. Cohen and Tinglong Dai, and part of the Springer Series in Supply Chain Management (edited by Prof. Chris Tang).

TEACHING

Current Teaching

Business Analytics <i>Esade Business School – Full-time MBA</i>	2023–present
Supply Chain Management <i>Esade Business School – Full-time MBA</i>	2023–present
Data Science and Operations for Sustainability <i>Esade Business School – MSc Sustainability</i>	2023–present
Simulation, Optimization and Computational Methods <i>Esade Business School – PhD Program</i>	2025–present

Invited Teaching

Supply Chain 4.0: AI, Data & Smart Decision-Making <i>Antwerp Management School, Antwerp, Belgium – Executive Masterclass</i>	2025–present
Data-driven Decision-Making and Business Analytics – <i>CLSBE, Lisbon, Portugal – Executive-MBA</i>	2022–present
Green Transport and Machine Learning <i>KU Leuven Kortrijk, Belgium – Postgraduate Smart Logistics</i>	2019–present

Past Teaching

Data-Driven Organizations <i>Esade Business School – Custom Program for GB Foods</i>	2025
Inventory Control and Machine Learning in Supply Chains <i>University of Antwerp, Belgium – Executive Masters Program</i>	2019–2023
Operations Management <i>CLSBE, Lisbon, Portugal – Executive Masters Program</i>	2021–2022
Supply Chain Management <i>CLSBE, Lisbon, Portugal – M.Sc. Business Analytics</i>	2021–2023
Advanced Predictive Analytics <i>CLSBE, Lisbon, Portugal – M.Sc. Business Analytics</i>	2022–2023
Operations Analytics <i>CLSBE, Lisbon, Portugal – M.Sc. Business Analytics</i>	2022–2023
Introduction to Operations Management <i>CLSBE, Lisbon, Portugal – B.Sc. Business Administration and Applied Economics</i>	2020–2023
Data-driven Decision Making and Business Analytics <i>CLSBE, Lisbon, Portugal – B.Sc. Business Administration and Applied Economics</i>	2022–2023
Deep Reinforcement Learning in Inventory Management <i>NYU Stern School of Business, US – PhD Program</i>	2022
Deep Reinforcement Learning in Inventory Management <i>Wharton, US – PhD Program</i>	2022
Impact of Machine Learning on Operations Management <i>Vlerick Business School, Belgium – Supply Chain Bootcamp</i>	2019
ML in Operations Management <i>Catholic University Milan, Italy – Workshop</i>	2019

SERVICE

Reviewer for *Management Science*, *Operations Research*, *Manufacturing Services & Operations Management*, *Production and Operations Management*, *European Journal of Operational Research*, *International Journal of Production Economics*, *Omega*, *INFORMS Journal on Computing*.